Chronic Kidney Disease – Diagnosis and Management

Effective Date: July 23, 2025

Scope

This guideline provides recommendations for the investigation, evaluation, and management of adults at risk of/or with known chronic kidney disease (CKD).

Key Recommendations

- Screen high-risk patients (e.g., diabetes, hypertension, cardiovascular disease, family history, and those with a history of acute kidney injury) using estimated Glomerular Filtration Rate (eGFR) and urine Albumin-Creatinine Ratio (ACR). Confirm abnormal test results with a repeat measurement and obtain urinalysis (microscopic).
- Determine likely cause of kidney disease where possible. This has important implications for determining risk of End Stage Kidney Disease (ESKD)/Kidney Failure and other complications.
- Use disease modifying drugs to control hypertension and proteinuria that prevent or postpone kidney function decline.
- **All patients with CKD:** Initiate an ACE inhibitor (ACE-I) or angiotensin receptor blocker (ARB).
- **[NEW] Patients with CKD and uACR ≥ 20 mg/mmol:** In addition to ACE-I/ARB, start a sodium-glucose co-transporter 2 inhibitor (SGLT2i), unless contraindicated.
- **[NEW] Patients with CKD and heart failure:** In addition to ACE-I/ARB, start a SGLT2i, unless contraindicated.
- **[NEW] Patients with CKD and type 2 diabetes:** In addition to ACE-I/ARB and SGLT2i, include a non-steroidal mineralocorticoid receptor antagonist (ns-MRA) and a glucagon-like peptide-1 receptor agonist (GLP-1 RA) to optimize renal and cardiovascular outcomes.
- Prescribe statins for CKD patients ≥ 50 yrs and for those 18-49 yrs with cardiovascular risk factors.
- Hold ACE-I, ARB, SGLT2i, and diuretics if patient has acute illness with dehydration, with a plan to restart.
- To assist in determining the need for and timing of referral, obtain advice from local internists, nephrologists or RACE.

Background

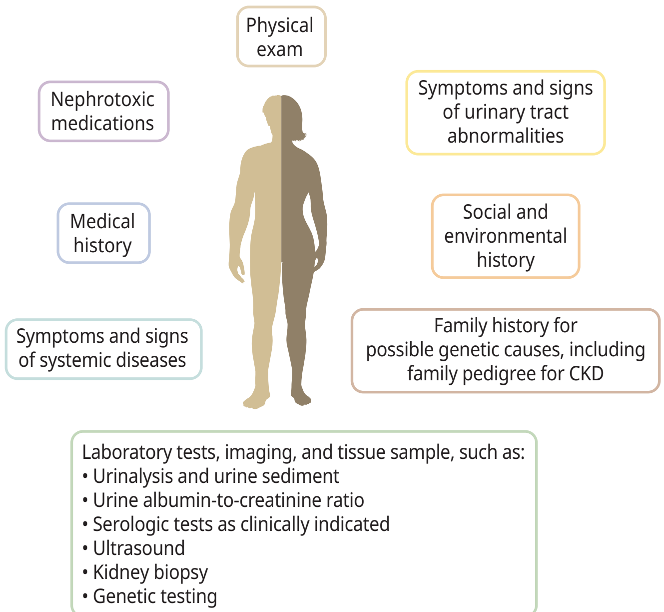
CKD is defined as abnormalities of kidney structure or function that are present for a minimum of 3 months, with implications for health.[1] CKD markedly increases the risk of adverse health outcomes, including cardiovascular mortality, atrial fibrillation, coronary heart disease, stroke, heart failure, peripheral artery disease, acute kidney injury, and kidney failure requiring replacement therapy (dialysis or transplantation).[2] CKD increases the risk of all-cause mortality.[2–4] Most patients with CKD die from other comorbidities before they progress to kidney failure.[1] Medications are now available that improve mortality and reduce major kidney disease events.

In 2022/23, over 240,000 British Columbians were identified as having CKD (4.1% of the adult population),[5] which is lower than the expected 10% of adult population.[6] This underscores the need for enhanced education about CKD identification and awareness.

Evaluation for the Cause(s) of CKD

A comprehensive approach (Figure 1) is recommended to establish the cause of CKD such as clinical context, personal and family history, social and environmental factors, medications, physical examination, laboratory tests, imaging, and genetic and pathologic diagnoses.[1,7] The two most common causes of CKD are hypertension and diabetes, which often co-exist.

Figure 1: Evaluation of the Cause(s) of Chronic Kidney Disease.
Adapted with permission, with minor edits, from KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease.



Even if a cause seems obvious (e.g., diabetes), the possibility of other diagnoses, including underlying primary kidney disorder (e.g., glomerulonephritis) should be considered in all patients, and especially those with:

- Abnormal urinalysis, (e.g., proteinuria, hematuria, cellular casts). Note: Hyaline casts are normal. See BCGuidelines: Workup of Microscopic Hematuria guideline
- Decline in eGFR > 10-15% per year despite correction of reversible precipitants (e.g., volume contraction, febrile illness, medications)
- Constitutional symptoms suggesting systemic illness (e.g., lupus nephritis, heart failure, HIV, liver disease, and dysproteinemias)
- Sudden onset or severe worsening of symptoms (e.g., edema unrelated to heart or liver disease)

Screening

Consider screening individuals with these common risk factors:

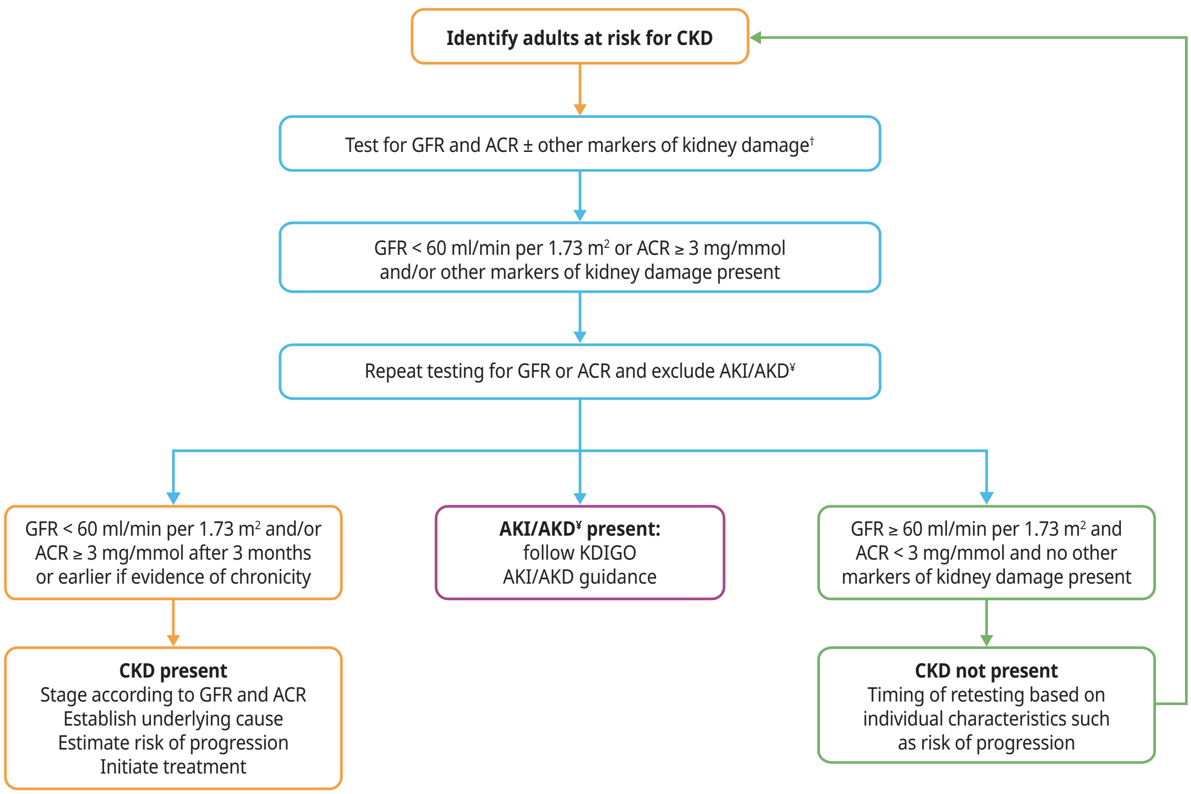
- First degree family history of kidney disease
- Diabetes
- Hypertension
- Cardiovascular disease
- Prior acute kidney injury

- Check eGFR and urine ACR once every 1-2 years, depending upon clinical circumstances (e.g., annual screening for those with diabetes, or who have had a cardiovascular event). Note: Older age alone is not a reason for screening.

Diagnosis

CKD is diagnosed when eGFR and/or urine ACR are abnormal and present for at least 3 months. Therefore, it cannot be diagnosed with **one isolated** abnormal measurement. Figure 2 outlines the recommended diagnostic and staging algorithm. Other markers of kidney damage include urine abnormalities (e.g., hematuria), structural abnormalities on imaging studies (e.g., polycystic kidneys) or histological findings on kidney biopsy.

Figure 2: Screening Algorithm for Diagnosis and Staging of CKD in Adults.
Adapted with permission, with minor edits, from KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease.



ACR - albumin-to-creatinine ratio; CKD - chronic kidney disease; GFR - glomerular filtration rate; AKI - Acute Kidney Injury. †Markers of kidney damage other than albuminuria may also be used to diagnose CKD, but albumin-to-creatinine ratio (ACR) and GFR are still required to determine stage and estimate risk of progression. ¥AKI is the presence of any of the following: increase in serum creatinine by > 26 μmol/L within 48 hours or > 1.5x the baseline or urine volume less than 0.5 mL/kg/h for at least 6 hours.

Estimated GFR (eGFR) Values and Interpretation

- Values of < 60 mL/min per 1.73 m^2 which are present for ≥ 3 months are diagnostic of CKD.
- For patients with a new or unexpected eGFR < 60 mL/min per 1.73 m^2 or a significant change in eGFR, repeat testing should be conducted within 1-2 weeks, depending on the severity and clinical context, to confirm stability or identify rapid deterioration.
- Initial values of eGFR between 50-60 mL/min per 1.73 m^2 and the absence of other markers of kidney disease (albuminuria, hematuria, structural abnormalities) may not indicate CKD, and need to be contextualized within trends over time.

Caveats of Interpreting eGFR

- eGFR is calculated from serum biomarkers such as creatinine. eGFR is an estimated value that assumes a steady state of the biomarker. All labs in BC automatically report eGFR when creatinine is ordered.
- In hospitalized patients or patients with AKI, fluctuations in creatinine make eGFR unreliable. In these circumstances, creatinine values and changes in creatinine (not eGFR) over time should be used to guide management.
- eGFR may be unreliable in extremes of muscle mass, while on certain diets (e.g., very high or very low protein), or when medications that interfere with the excretion of creatinine are used (e.g., trimethoprim, fenofibrate).[1] More accurate markers (e.g., Cystatin C) are being introduced in clinical care, however, not currently available in BC.
- As a general rule, eGFR can be used to guide outpatient drug dosing even if the references use creatinine clearance.
- Please discuss with a Nephrologist or contact RACE when accuracy of GFR may affect clinical decision-making (e.g., nephrotoxic drugs).

Urine ACR Values and Interpretation

- Urine ACR is the preferred method to screen for protein in the urine.
  - Urine ACR elevation (≥ 3.0 mg/mmol) on serial testing (≥ 2 elevated results out of 3 tests over a 4- to 12-week period) is abnormal.
  - In most cases, 24-hour urine collections are not necessary to assess protein excretion.
  - If laboratory testing of urine ACR and protein/creatinine ratios is unavailable, dipstick protein measurements can be performed.
- Urine ACR may be transiently elevated in some patients due to acute illness, vigorous exercise, poorly controlled hypertension or poorly controlled blood glucose. Repeat testing should be done after correcting the transient factor, using a first morning urine sample, if possible, to ensure a concentrated urine that better reflects the albumin/creatinine ratio.
- See Appendix A for interpretation of ACR values.

Urinalysis (including urine microscopy)

- Significant abnormalities include persistent red blood cells (RBC) or white blood cells (WBC) in the absence of infection or instrumentation, and the presence of cellular casts.
- Persistent urine test abnormalities, (presence of WBCs, RBCs, non-hyaline casts, and/or protein or albumin) even with eGFR values ≥ 60 mL/min per 1.73 m^2 suggest kidney disease.
- Urine microscopy should be collected at the laboratory as the sample must be analysed within 2-3 hours. See BCGuidelines: Workup of Microscopic Hematuria.

Imaging

Renal imaging (kidney and bladder) is necessary to assess for structural abnormalities and aid in determining the etiology of CKD. Renal ultrasound (kidney and bladder) is a reliable test and typically most accessible. CT or MRI may provide more detailed information, if needed.

- Ultrasound should be performed or repeated in the following cases:
  - Obstructive urinary symptoms or those with suspicion of benign prostatic hypertrophy
  - Unexplained microscopic or macroscopic hematuria
  - Family history of structural kidney disease (e.g., polycystic kidney disease)

Classification and Staging of CKD

Figure 3 summarizes the classification and staging of CKD based on CGA: Cause (C), eGFR value (G1–G5) (G), and degree of albuminuria (A1–A3) (A). CGA classification also outlines the recommended frequency of eGFR and urine ACR monitoring. Staging is important for care planning and medical management. Patient age, ethnicity, disease cause, current eGFR and proteinuria levels also may impact outcomes.

Figure 3: Classification of CKD and Frequency of Monitoring.

Adapted with permission, with minor edits, from KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease

CKD is classified based on: • Cause (C) • GFR (G) • Albuminuria (A)				**Albuminuria categories** Description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				< 3 mg/mmol	3–29 mg/mmol	≥ 30 mg/mmol
GFR categories (ml/min/1.73 m²) Description and range	**G1**	Normal or high	≥ 90	Screen 1	Treat 1	Treat 3
	G2	Mildly decreased	60–89	Screen 1	Treat 1	Treat 3
	G3	Mildly to moderately decreased	45–59	Treat 1	Treat 2	Treat 3
	G4	Moderately to severely decreased	30–44	Treat 2	Treat 3	Treat 3
	G5	Severely decreased	15–29	Treat 3	Treat 3	Treat 4+
	G6	Kidney failure	< 15	Treat 4+	Treat 4+	Treat 4+

Low risk (if no other markers of kidney disease, no CKD)

Moderately increased risk

High risk

Very high risk

A1-A3 and G1-G5 is used to classify CKD. The numbers in the boxes, 1-4 refer to the frequency of lab test monitoring per year. The colors indicate risk for CKD progression. CKD - chronic kidney disease; GFR - glomerular filtration rate.

Risk Calculators for Kidney Failure and CV events

The Kidney Failure Risk Equation (KFRE) is recommended to estimate a patient's 2- and 5-year risk of progression to kidney failure requiring dialysis or transplantation, in the absence of intervention. The basic KFRE uses 4 variables (age, sex, eGFR, urine ACR) to calculate risk. It is validated in those with eGFR < 60 ml/min per 1.73 m^2. For adults with eGFR > 60 mL/min per 1.73 m^2, there are other risk prediction models using eGFR and urine ACR. **The risk is modifiable.** Based on the risk of progression to kidney failure, patients may be managed in primary care or prioritized for referral. See Figure 4 showing eGFR and KFRE risk criteria and recommendations for referral and treatment planning.

Figure 4: Risk Based Approach to CKD Care.
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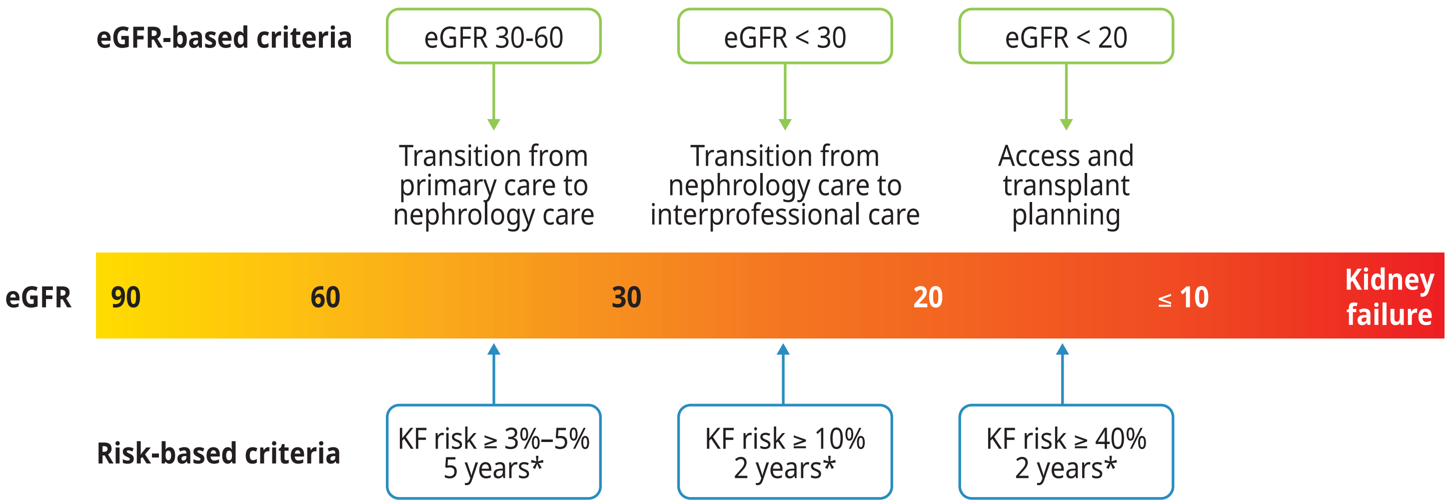


* risk for progression to kidney failure
eGFR - estimated glomerular filtration rate, KF - Kidney failure

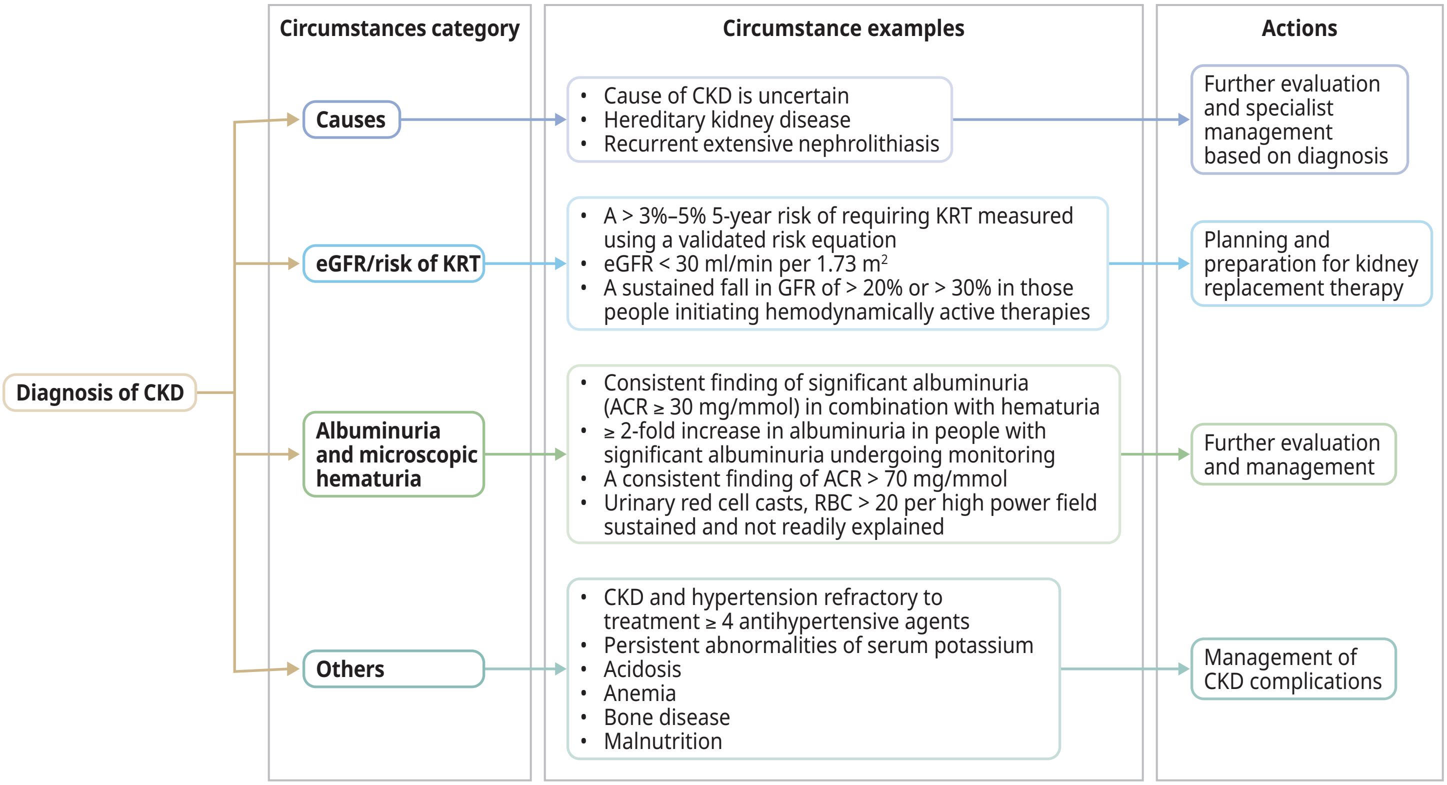
For cardiovascular risk prediction, use of validated tools developed specifically for adults with CKD such as QRISK3, PREVENT is recommended. Further details on risk determination are available on the Kidney Disease Improving Global Outcomes (KDIGO) CKD management guideline.

Management

Many instances of newly identified CKD can be followed and managed in primary care but sometimes, early specialist collaboration is needed. Figure 5 outlines some clinical scenarios requiring specialist referral.

The cause of kidney disease (e.g., polycystic kidney disease, glomerulonephritis, diabetes) can affect the rate at which kidney disease worsens. Management of hypertension, proteinuria, and use of disease modifying drugs can prevent or postpone kidney function decline.[1] This underlines the importance of early detection, evaluation, changes to health behaviors, and appropriate management of individuals with kidney disease.

Figure 5: Circumstances for Referral to Specialist Kidney Care Services and Goals of Referral.
Adapted with permission, with minor edits, from KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease.



ACR - albumin-to-creatinine ratio; AER - albumin excretion rate; CKD - chronic kidney disease; eGFR - estimated glomerular filtration rate; KRT - kidney replacement therapy; PCR - protein creatinine ratio; PER - protein excretion rate; RBC - red blood cells.

Indications for Urgent and Priority Referrals

Urgent referral indicated (urgent concerns should trigger a conversation with nephrologist):

- Presence of active urine sediments (red blood cell casts or cellular casts ± protein), especially when associated with reduced eGFR
- AKI in absence of readily reversible cause (e.g., volume depletion, NSAIDs)
- Abrupt, sustained fall in eGFR
- eGFR < 15 mL/min per 1.73 m^2
- Nephrotic syndrome (> 300 mg/mmol urine protein-to-creatinine ratio, < 30 g/L serum albumin, edema, hyperlipidemia, lipiduria)

Priority referral indicated (patient to be seen in a timely fashion):

- eGFR < 30 mL/min per 1.73 m^2
- Unexplained persistent urine ACR > 30-70 mg/mmol, regardless of eGFR (e.g., in absence of diabetes or HTN)
- Progressive CKD, with eGFR decline > 5 mL/min per 1.73 m^2 /year
- Diabetes and evidence of CKD with eGFR < 45 mL/min per 1.73 m^2, urine ACR > 30 mg/mmol

Recommended items to be included in referral package:

- Comorbidities (especially cardiovascular)
- Medications
- CBC
- Electrolytes (Na, K, Cl, HCO_3), calcium
- Creatinine/eGFR (include current value and any historical values available)
- Urinalysis and urine microscopy
- Urine ACR
- Kidney ultrasound – not required prior to submitting referral, but recommend it be arranged with result sent to specialist when completed

Referral Resources:

- Use PathwaysBC to see the list of specialists and their wait times.
- Real-time communication with local specialists (or RACE) can provide rapid advice for urgent cases and timely advice and facilitate the most appropriate mechanism of referral.
- To locate kidney clinics in the region, use the BC Renal website.

A urology consult is more appropriate than nephrology in the following scenarios:

- Kidney mass
- Gross hematuria
- Enlarged prostate
- Obstruction
- Large symptomatic or obstructing kidney stone

Non-pharmacological Management

Patients with CKD require a comprehensive treatment strategy to reduce risks of progression and associated complications. They benefit from multidisciplinary teams who can support them to make beneficial changes. This may include addressing diet, smoking and physical activity, and understanding of medications. It may also help address the uncertainty and emotional distress.

Evidence suggest an important role of diet, including avoidance of excessive sodium and protein intake, which can cause or worsen glomerular hyperfiltration.[8] Refer to diet information sheets provided by BC Renal intended for use by kidney patients. Consider consulting a renal dietitian through BC Dieticians.

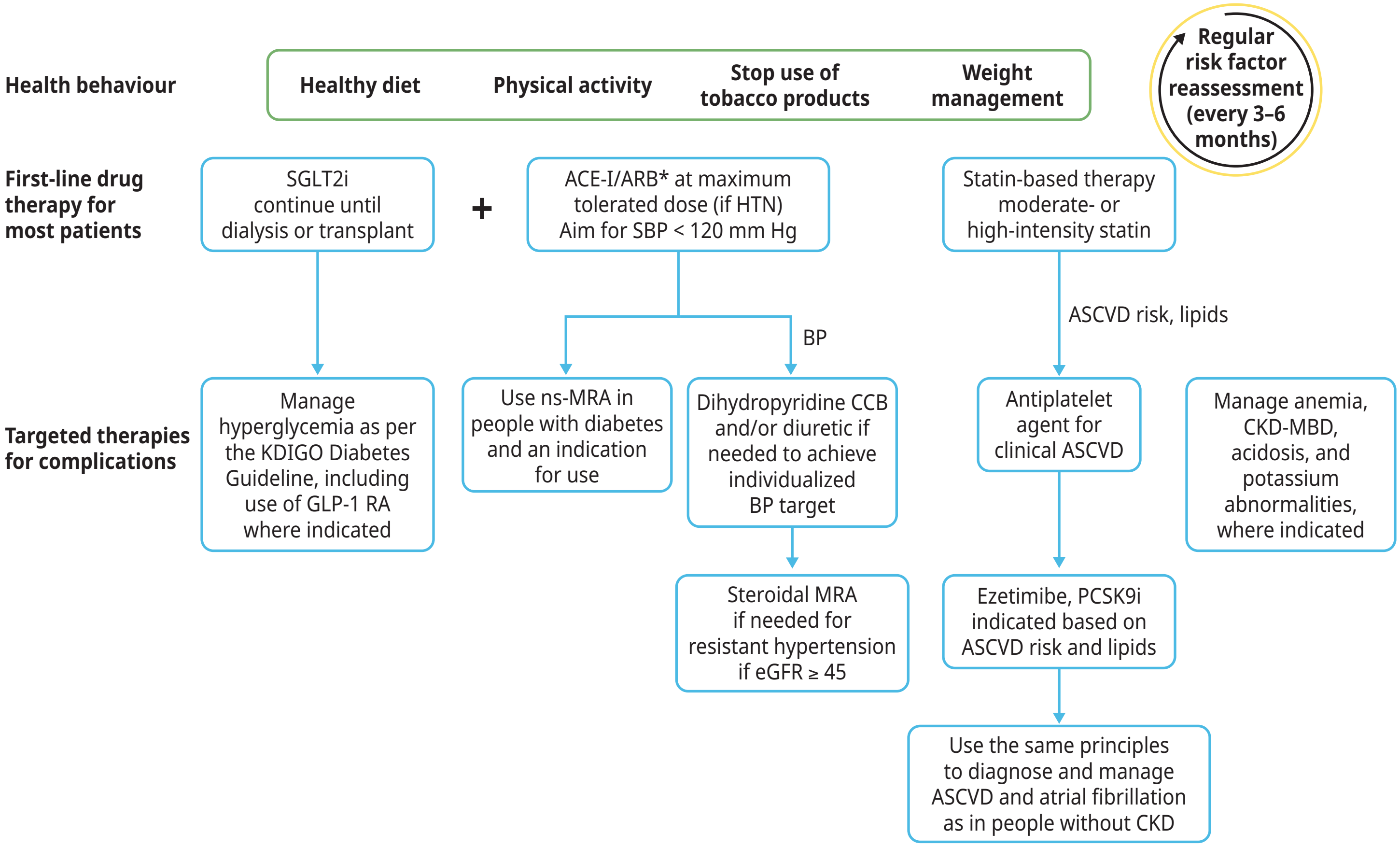
Patients with CKD should be encouraged to engage in physical activities in line with their health and capacity, aiming to meet the KDIGO guideline recommendation of 150 minutes of moderate to vigorous aerobic (cardio) activity per week.[1]

Support patient self-management with the means available in your community. Promotion of healthy lifestyles can prevent CKD in people at risk and help delay progression.

Pharmacological Management

Figure 6 outlines an approach to pharmacological treatments commonly used in patients with CKD, including targeted therapies for specific conditions, such as diabetes. More details and cautions on management of these co-morbidities are in Table 1.

Figure 6: Approach to Chronic Kidney Disease Treatment.
Adapted with permission, with minor edits, from KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease.



* Angiotensin-converting enzyme inhibitor or angiotensin II receptor blocker should be first-line therapy for blood pressure (BP) control when albuminuria is present; otherwise dihydropyridine calcium channel blocker (CCB) or diuretic can also be considered.

ASCVD - atherosclerotic cardiovascular disease; CKD-MBD - chronic kidney disease-mineral and bone disorder; eGFR - estimated glomerular filtration rate; GLP-1 RA - glucagon-like peptide-1 receptor agonist; HTN - hypertension; KDIGO - Kidney Disease: Improving Global Outcomes; MRA - mineralocorticoid receptor antagonist; ns-MRA, nonsteroidal mineralocorticoid receptor antagonist; PCSK9i - proprotein convertase subtilisin/kexin type 9 inhibitor; SBP - systolic blood pressure; SGLT2i - sodium-glucose cotransporter-2 inhibitor.

Table 1: Care Objectives and Targets for CKD

Domain	Recommendations
Diabetes	• Long-acting sulfonylureas are associated with hypoglycemia and are not recommended. • SGLT2i improves glycemic control (if eGFR > 45) and reduces the risks of CVD and CKD progression in patients with diabetes. • Semaglutide significantly reduces the risk of major kidney disease events in those with type 2 diabetes[9] • In those with diabetes and ongoing proteinuria, ns-MRA is recommended. • In those with unstable eGFR or acute changes in clinical condition, metformin should be held, dose-reduced or avoided (see Appendix C). Risk/benefit of metformin can be challenging and requires discussion with a specialist if eGFR < 30. • HbA1C ≤ 7.0% may not be appropriate for older frail patients since it may lead to hypoglycemia.
Blood Pressure	• ACE-I or ARB is generally first line therapy in those with CKD with CVD risk, especially those with proteinuria. • Measure and record BP at diagnosis and at every visit thereafter. • Ambulatory BP monitoring is encouraged. • BP targets for CKD vary by guideline however most would recommend a target of 120/80. Use clinical judgment when interpreting BP targets. Consider comorbidities and prognosis. • Consultation with a specialist is recommended for complex patients, especially elderly patients with multiple comorbidities.
Avoidance of Acute Kidney Injury	• Hold ACE-I, ARB, SGLT2i, and diuretics if patient has acute illness with dehydration with a plan to restart. Refer to BC Renal Sick Day Hold List. • SGLT2i and diuretics should be held if having surgery. • The above medications can be restarted safely if there is no evidence of AKI. AKI is the presence of any of the following: increase in serum creatinine by > 26 μmol/L within 48 hours or > 1.5x the baseline or urine volume less than 0.5 mL/kg/h for at least 6 hours. • If contrast dye (especially arterial) is required, the risk can be mitigated with pre-hydration. The risk of contrast-induced AKI is higher in people with volume depletion. Talk to a nephrologist or use a published protocol. Creatinine should be measured pre and post procedure within 3 days. • Transient insults to the kidney may result in a change in trajectory of stable kidney function.
Medications	• Drug interactions are common in CKD and most are avoidable. • In CKD, some medications need to be used with caution or dose-adjusted for the level of eGFR. • Medication review is critical both during and after an episode of AKI. • If hospitalized or changing serum creatinine, seek additional advice regarding drug dosing. Refer to BC Renal Agency Pharmacy & Formulary information.
Kidney Function Measurements	• Refer to Figure 3 for frequency of eGFR and urine ACR measurements. • Check BP, serum creatinine, and serum potassium within 2-4 weeks after initiation or dose increase of medications known to impact renal function (e.g., ACE, ARB and MRAs). See Appendix C for information on CKD drugs.
CVD Risk Assessment	• Calculate and record CVD risk. Use tools such as PREVENT and QRISK3. • Manage in accordance with relevant guidelines. • In patients with newly identified CKD check lipid profile. • Treat adults aged ≥ 50 years with CKD and eGFR ≤ 60 (GFR categories G1–G2) with a statin. • Choose statin-based regimens to maximize the absolute reduction in LDL to achieve the largest treatment benefits. Afterwards, monitoring of lipids is not necessary.
Vaccinations	• Influenza, Pneumococcal, COVID vaccines are recommended in all CKD patients. • Refer to Immunizebc.ca for other recommended vaccines (e.g., Hepatitis B, MMR, Varicella).

Special Populations

Elderly and Frail

Older adults make up the largest group with advanced CKD and require individualized care. Creatinine-based eGFR may overestimate kidney function due to lower muscle mass, increasing the risk of drug overdosing—consider alternative assessments and adjust medications accordingly. Less intensive BP and A1C targets are often necessary to reduce falls, hypotension and hypoglycemia. For those with frailty or low muscle mass, higher protein and calorie intake may be needed; refer to a renal dietitian for guidance.

Pregnancy and Reproductive Health

Although CKD is associated with decreased fertility, pregnancy is possible at all stages of CKD.[1] People with CKD are at risk for adverse pregnancy-associated outcomes, including progression of their underlying CKD, a flare of their kidney disease, and pregnancy complications including pre-eclampsia, preterm delivery, and small for gestational age infant.[1] Those who have pre-eclampsia are at increased risk for CVD and CKD later in life, and should be assessed for both. All persons with CKD of reproductive age should receive counselling with respect to safe and effective contraception and options/optimization for pregnancy, if desired.

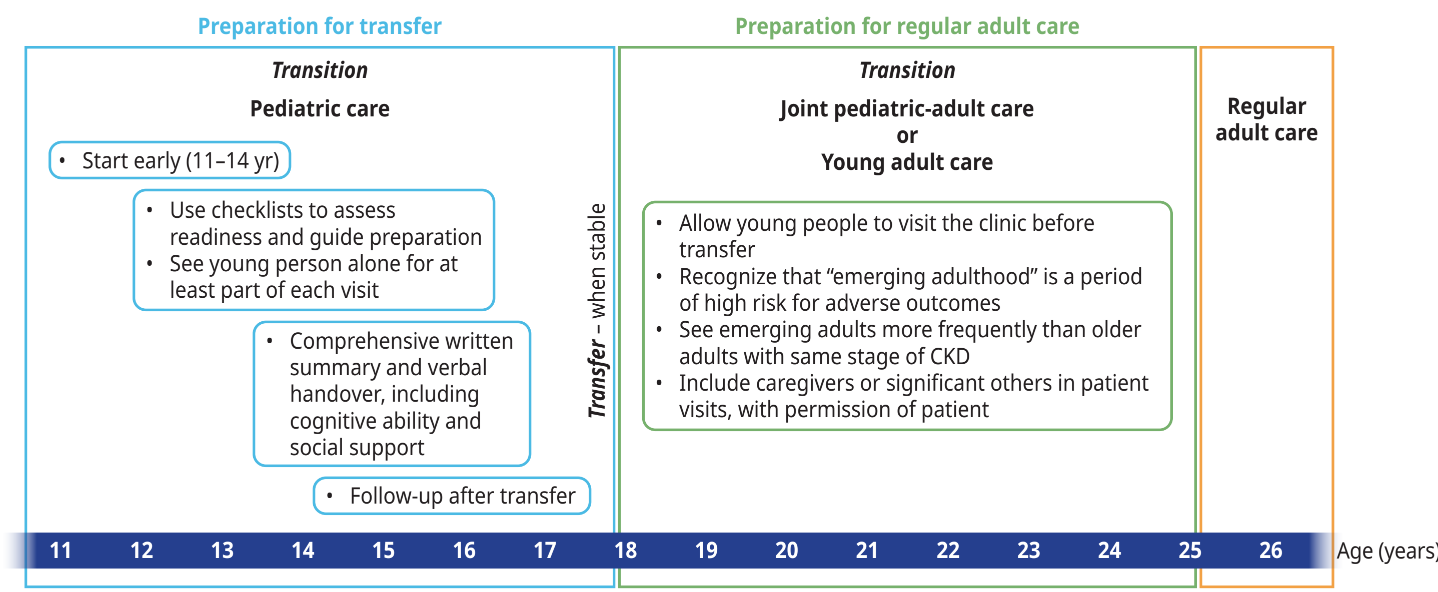
Transgender Population

The existing eGFR equations use sex as a binary variable. Therefore, in people who are transgender, gender-diverse, or nonbinary, where a person's gender identity is different from their sex assigned at birth, and/or they are taking gender affirming hormone therapy, the eGFR may be less accurate.[1] Conversation with a nephrologist or a laboratory specialist is recommended.

Transition of Care for Pediatric Patients into Adult Care

Coordination with pediatric nephrology to enable transition over a period of 12-18 months minimum is recommended. Start preparing patient at 11-14 yrs of age for transfer to adult-oriented care.

Figure 7: Transition from Pediatric to Adult Care in CKD.
Adapted with permission, with minor edits, from KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease.



CKD - chronic kidney disease

Indigenous Populations

Indigenous people are at higher risk of CKD, irrespective of diabetic status, due to other multiple risk factors. These include biological reasons (e.g., inherited diseases, systemic inflammatory conditions, premature birth, small for gestational age), environmental (e.g., air pollution, water pollution), and social determinants of health.[10–13] A thorough evaluation of all potential causes of CKD is warranted.

Advance Care Planning

Encourage patients with CKD to consider advance care planning and their goals for future care, including for the end of life. Clinicians can find resources to help support difficult discussions, such as choosing not to start dialysis (conservative care pathway) or stopping dialysis treatments on the BC Renal Agency Palliative Care. Other resources for palliative care can also be found at the Provincial Advance Care Planning and the BC Guidelines on Palliative Care.

Methodology

This guideline is an updated version of the BCGuidelines: Chronic Kidney Disease guideline (2019). The recommendations are tailored to support primary care practice in BC and are based on recommendations from Kidney Disease: Improving Global Outcomes (KDIGO) 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. In situations where there is a lack of rigorous evidence, the experience-informed clinical opinion is provided to support decision making and high-quality patient care. The guideline development process included engagement and consultation with primary care providers, specialists and key stakeholders.

Resources

Abbreviations

ACE-I	Angiotensin Converting Enzyme Inhibitors
ACR	Albumin-to-Creatinine Ratio
AKI	Acute Kidney Injury
ARB	Angiotensin Receptor Blocker
CGA	Cause, eGFR, and Albuminuria (CKD classification)
CKD	Chronic Kidney Disease
CVD	Cardiovascular Disease
eGFR	Estimated Glomerular Filtration Rate
GLP-1 RA	Glucagon-like Peptide-1 Receptor Agonist
KDIGO	Kidney Disease: Improving Global Outcomes
MRA	Mineralocorticoid Receptor Antagonist
ns-MRA	Nonsteroidal Mineralocorticoid Receptor Antagonist
NSAIDs	Nonsteroidal Anti-inflammatory Drugs
PCSK9i	Proprotein Convertase Subtilisin/Kexin Type 9 Inhibitor
SGLT2i	Sodium-Glucose Cotransporter-2 Inhibitors

Practitioner Resources

- **BC Renal:** Clinical rresources for physicians, dieticians, pharmacists and information for patients. See bcrenal.ca
- **Rapid Access to Consultative Expertise (RACE):** Rapid Access to Consultative Expertise Program allows Physicians (SPs and FPs), Medical Residents, Nurse Practitioners and Midwives to go to one online application or call one number and speak directly to specialists. See raceconnect.ca.
- **Real Time Virtual Support (RTVS):** Available to support practitioners in rural, remote, and First Nations communities in BC.
  - Rural Urgent Doctor in aid (RUDi) for instant emergency medicine support.
  - Child Health Advice in Real-Time Electronically (CHARLiE) for instant pediatric support.
- **Provincial Laboratory Medicine Services (PLMS):** Accountable for various functions related to the administration of lab services in B.C., and for implementing policies and processes on behalf of the Ministry of Health. To add a test to an existing order, change a test priority or to obtain a test result by phone, see PLMS Contact Us page, call 604-714-2800 or email plmsinfo@phsa.ca.
- **PathwaysBC:** An online resource that allows physicians, nurse practitioners and their office staff to quickly access current and accurate referral information. This includes specialists and specialty clinic wait times and areas of expertise. See: pathwaysbc.ca/login
- **Health Data Coalition:** An online, physician-led data sharing platform that can assist you in assessing your own practice in areas such as chronic disease management or medication prescribing. HDC data can graphically represent patients in your practice with chronic diseases in a clear and simple fashion, allowing for reflection on practice and tracking improvements over time. See: Health Data Coalition – Better Information. Better Care. Better Patient Outcomes. (hdcbc.ca)
- **Family Practice Services Committee**
  - Practice Support Program: Offers focused, accredited training sessions for BC physicians to help them improve practice efficiency and support enhanced patient care.
  - Chronic Disease Management and Complex Care Incentives: Compensates family physicians for the time and skill needed to work with patients with complex conditions or specific chronic diseases.
- **Relevant BC Guidelines**
  - Cardiovascular Disease - Primary Prevention
  - Diabetes Care
  - High-Risk Drinking and Alcohol Use Disorder
  - Hypertension - Diagnosis and Management
  - Tobacco Use Disorder (TUD)
  - Ultrasound Prioritization
  - Workup of Microscopic Hematuria

Patient, Family and Caregiver Resources

- **HealthLinkBC:** Patients can call HealthLinkBC at 8-1-1 toll-free in B.C., or for the deaf and the hard of hearing, call 7-1-1. They will be connected with an English-speaking health-service navigator, who can provide health and health-service information and connect them with a registered dietitian, exercise physiologist, nurse, or pharmacist. See: healthlinkbc.ca/
  - Chronic Kidney Disease | HealthLink BC
- **BC Renal:**
  - Self-Management
  - Diet
  - Sick Day Medication Hold List
  - Kidney Care (Non-Dialysis)
- Kidney Wellness Hub
  - Staying Active
- **Advance Care Planning:** Making Future Health Care Decisions. See www2.gov.bc.ca/gov/content/family-social-supports/seniors/health-safety/advance-care-planning

Diagnostic Codes

585 - Chronic Renal Failure

Appendices

- Appendix A: Interpretation of Urine ACR Values to Assess Albuminuria and Proteinuria
- Appendix B: Recommended Drug Modifications in Presence of Acute Kidney Injury (AKI)
- Appendix C: Medications for Chronic Kidney Disease

Associated Documents

The following document accompanies this guideline (available on the website):

- List of Contributors

References

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BC Guidelines are developed for the Medical Services Commission by the Guidelines and Protocols Advisory Committee, a joint committee of Government and the Doctors of BC. BC Guidelines are adopted under the *Medicare Protection Act* and, where relevant, the *Laboratory Services Act*.

Disclaimer: This guideline is based on best available scientific evidence and clinical expertise as of July 23, 2025. It is not intended as a substitute for the clinical or professional judgment of a health care practitioner.

Appendix A: Interpretation of Urine ACR Values to Assess Albuminuria and Proteinuria

Categories	Urine ACR (mg/mmol)	Protein reagent strip measuring total protein	PCR urine protein-to-creatinine ratio (mg/mmol)
Normal to mildly increased	< 3.0	Negative to trace	< 15.0
Moderately increased	3.0 – 30.0	Trace to +	15.0 – 50.0
Severely increased	> 30.0	+ or greater	> 50.0

Appendix B: Recommended Drug Modifications in Presence of CKD and in Acute Kidney Injury (AKI)*

This list is not inclusive and only provides some examples of drugs that require adjustment in CKD and AKI. Primary care practitioners should consult a pharmacist or specialist for more information about specific medication questions. Refer to Appendix C for specific information pertaining to drugs used in CKD.

Medication	Pathophysiology	Recommendation in CKD	Hold in AKI?
NSAIDS (e.g., ibuprofen)	Decreases renal perfusion, interstitial nephritis, analgesic nephropathy	Use with caution in CKD. Consider alternatives (e.g., acetaminophen). Hold temporarily during times of prolonged fasting, critical illness, acute management of adverse effects, or 48-72 hours prior to elective surgery.	Yes
ACE-I (e.g., ramipril), ARB (e.g., losartan)	Protective in proteinuric CKD, diabetes, and heart failure but can cause decreased renal perfusion and hyperkalemia	Hold temporarily during times of prolonged fasting, critical illness, acute management of adverse effects, or 48-72 hours prior to elective surgery, Hold temporarily 24-48 hrs before and 48 hours after radiocontrast exposure in people with AKI or eGFR < 30 ml/min. A plan for recommencement is essential.	Yes
Potassium sparing diuretics (e.g., spironolactone, eplerenone)	Volume depletion and hyperkalemia	In CKD (other than in AKI), dose and frequent monitoring is essential if eGFR < 50 mL/min.	Yes
Metformin	Increased risk of metformin associated lactic acidosis	Avoid if eGFR < 30 mL/min. Hold temporarily 48-72 hours prior to elective surgery or during acute management of adverse effects. Hold temporarily 24-48 hours before and 48 hours after radiocontrast exposure in people with AKI or eGFR < 30 ml/min. Hold temporarily at the time of/before and 48 hours after iodinated contrast media injection in people with AKI.	Yes
SGLT2i (e.g., dapagliflozin)	Protective in diabetic nephropathy and cardiovascular disease but can cause decreased renal perfusion in the setting of volume depletion	Hold temporarily during times of prolonged fasting, critical illness, acute management of adverse effects, or 48-72 hours prior to elective surgery.	Yes
Diuretics (e.g., furosemide and hydrochlorothiazide)	Volume depletion and electrolyte abnormalities	Hold temporarily during times of prolonged fasting, critical illness, acute management of adverse effects, or 48-72 hours prior to surgery.	Yes, unless volume overloaded

Medication	Pathophysiology	Recommendation in CKD	Hold in AKI?
Opioids (e.g., hydromorphone, fentanyl, methadone)	Metabolites can accumulate	Reduce dose in CKD. For opioids that are considered safer in CKD, and opioids to avoid in CKD, consult: BCRenal - Preferred Medications in Chronic Kidney Disease.	Consider dose reduction
Pregabalin and gabapentin	Accumulation	Reduce dose and monitor for adverse effects. Avoid extended-release formulations in severe kidney dysfunction.	Consider dose reduction
Digoxin	Hyperkalemia Accumulation with side effects (bradycardia, confusion)	Reduce dose in CKD and monitor potassium and drug levels. Consider alternative therapy in the setting of kidney failure.	Adjust dose
Acyclovir/ valacyclovir	Risk of crystal nephropathy Drug accumulation and side effects (risk of seizures/confusion)	Reduce dose in CKD. Encourage hydration.	No. Ensure hydration
Statins (e.g., atorvastatin, rosuvastatin)	Risk of rhabdomyolysis	Consider dose reduction in CKD. Hold if rhabdomyolysis or unexplained/persistent muscle pain.	No
Phenytoin	Risk of accumulation and toxicity	Monitor levels and adjust level for serum albumin.	No
Lithium	Accumulation and increased risk of side effects Risk of nephrogenic diabetes insipidus Risk of chronic interstitial nephritis	Monitor lithium and electrolyte levels. Encourage hydration. Refer to a nephrologist if eGFR declines.	No
Hypoglycemics • Sulfonylureas (e.g., glyburide) • Meglitinides (e.g., Repaglinide) • Thiazolidinediones (e.g., pioglitazone)	Accumulation can increase the risk of hypoglycemia	Avoid long-acting preparations in moderate-severe CKD. May require dose adjustments.	No
Trimethoprim and co-trimoxazole	Risk of hyperkalemia Reduces tubular secretion of creatinine so leads to a rise in serum creatinine without renal injury	Reduce dose when eGFR < 30 mL/min, monitor eGFR and potassium.	No
Colchicine	Risk of accumulation and serious toxicity (GI, CNS)	Use a lower dose and consider other agents (e.g., corticosteroids).	No
Proton pump inhibitors	Risk of interstitial nephritis and tubulointerstitial nephritis	Clarify indication and consider another agent (e.g., H2 blocker).	No
Warfarin	Glomerular hemorrhage, kidney tubular damage, direct effects on kidney vascular calcification	Monitor INR and adjust dose as needed. Consider alternative (e.g., non-vitamin K antagonist oral anticoagulants).	No

For more information on dosing of DOACs in kidney disease please visit the 2018 European Heart Rhythm Association Guidelines (Figure 4).

For more information on common prescribing questions for patients with CKD, consult: Common Prescribing Questions for Patients with Chronic Kidney Disease not on Dialysis.

* Adapted from: *Acute Kidney Injury – potentially problematic drugs and actions to take in primary care*. “Think Kidneys” initiative by the UK Renal Registry in partnership with NHS England.

Appendix C: Medications for Chronic Kidney Disease

<table>
<tr><th>Generic Name
Trade name
Dosage form and strengths</th><th>Recommended Renoprotective Dose[A]</th><th>Approx. Cost per year[D]</th><th>PharmaCare Coverage[C]</th><th>Adverse Effects[B]</th><th>Drug Interactions[B] and Therapeutic Considerations for Chronic Kidney Disease</th></tr>
<tr><td colspan="6">Drugs for Chronic Kidney Disease</td></tr>
<tr><td colspan="6">Sodium-glucose Cotransporter-2 inhibitors (SGLT2i)
Examples are listed below. See BCGuidelines: Diabetes Care Antihyperglycemic drug table for more info.</td></tr>
<tr><td>Dapagliflozin
Forxiga, G
Tabs: 5, 10 mg</td><td>10 mg PO once daily (CKD trial dose)[1]</td><td>$270</td><td>Regular benefit</td><td rowspan="3">• Increased risk of genital mycotic infections; most can be managed with topical antifungal[4]
• Canagliflozin: decreased bone mineral density and increased risk of bone fractures
• Rare:
- Diabetic ketoacidosis
- Hypoglycemia (in absence of other hypoglycemics)</td><td rowspan="3">• Initiation or use does not necessitate alteration of frequency of CKD monitoring.[5]
• Reversible decrease in eGFR (< 30%) on initiation is generally not an indication to discontinue.[5]
• Reasonable to continue when the eGFR falls below 20 ml/min per $1.73m^2$, unless it is not tolerated or until kidney replacement therapy is initiated.[5]
• Withhold during times of prolonged fasting, critical illness, acute management of adverse effects, or 48-72 hours prior to surgery; have a plan to restart.[5]</td></tr>
<tr><td>Empagliflozin
Jardiance
Tabs: 10, 25 mg</td><td>10 mg PO once daily (CKD trial dose)[2]</td><td>$1,090</td><td>Limited coverage for treatment of type 2 diabetes mellitus (Special Authority)</td></tr>
<tr><td>Canagliflozin
Invokana
Tabs: 100, 300 mg</td><td>100 mg PO once daily (CKD trial dose)[3]</td><td>$1,140</td><td>Non-benefit</td></tr>
</table>

<table>
<tr><th>Generic Name
Trade name
Dosage form and strengths</th><th>Recommended Renoprotective Dose[A]</th><th>Approx. Cost per year[D]</th><th>PharmaCare Coverage[C]</th><th>Adverse Effects[B]</th><th>Drug Interactions[B] and Therapeutic Considerations for Chronic Kidney Disease</th></tr>
<tr><td colspan="6">Angiotensin Converting Enzyme Inhibitors (ACE-I)
Examples listed below. See BCGuidelines: Hypertension Antihypertensive drug table for more info.</td></tr>
<tr><td>Ramipril
Altace, G
Caps: 1.25, 2.5, 5, 10, 15 mg</td><td>Initial: 2.5 mg PO once daily
Usual: 2.5 - 10 mg PO once daily
Max: 20 mg PO per day</td><td>$80</td><td>Regular benefit (Reference Drug Program (RDP))

Non-benefit: 15 mg caps</td><td rowspan="2">Common
• Dry cough (8-12%)
• Hyperkalemia

Less Common
• Angioedema
• Precipitation of kidney failure in patients with renovascular disease, volume depletion or concomitant NSAID use</td><td rowspan="5">• Use highest approved dose that is tolerated for blood pressure control.[5]
• Avoid any combination of ACE-I, ARB, and direct renin inhibitor in people with CKD, with or without diabetes.[5]
• Check blood pressure, serum creatinine, and serum potassium within 2-4 weeks after initiation or dose increase.[5]
• Hyperkalemia can often be managed by measures to reduce serum potassium levels rather than decreasing or stopping treatment (refer to “Potassium Abnormalities” section below).[5]
• Continue unless serum creatinine rises by more than 30% within 4 weeks following initiation of treatment or an increase in dose.[5]
• Reasonable to continue even when the eGFR falls below 30ml/min per $1.73m^2$.[5]
• Consider dose reduction or discontinuation upon either symptomatic hypotension or uncontrolled hyperkalemia despite medical treatment, or to reduce uremic symptoms while treating kidney failure.[5]
• Withhold during times of prolonged fasting, critical illness, acute management of adverse effects, or 48-72 hours prior to surgery; have a plan to restart.[5]</td></tr>
<tr><td>Perindopril
Coversyl, G
Tabs: 2, 4, 8 mg</td><td>Initial: 4 mg PO once daily
Usual: 4 - 8 mg PO once daily
Max: 8 mg PO per day</td><td>$100</td><td>Partial benefit, RDP</td></tr>
<tr><td colspan="5">Angiotensin Receptor Blocker (ARB)
Examples listed below. See BCGuidelines: Hypertension Antihypertensive drug table for more info.</td></tr>
<tr><td>Losartan
Cozaar, G
Tabs: 25, 50, 100 mg</td><td>Initial: 25-50 mg PO once daily
Usual: 50 - 100 mg PO once daily (CKD trial dose)[6]</td><td>$100</td><td>Limited coverage for intolerance to ACE-I</td><td rowspan="2">Common:
• Hyperkalemia

Less Common:
• Angioedema
• Precipitation of kidney failure in patients with renovascular disease, volume depletion or concomitant NSAID use</td></tr>
<tr><td>Irbesartan
Avapro, G
Tabs: 75, 150, 300 mg</td><td>Initial: 75 - 150 mg PO once daily
Usual: 150 - 300 mg PO once daily
Max: 300 mg PO per day (CKD trial dose)[7]</td><td>$90</td><td>Limited coverage for intolerance to ACE-I and failure on candesartan, losartan, telmisartan</td></tr>
<tr><td colspan="6">HMG-CoA Reductase Inhibitors (Statins) – Moderate or High-Intensity
See BCGuidelines: Cardiovascular Disease Statins drug table for more info.</td></tr>
<tr><td colspan="6">Large trials have shown the following once-daily intensive statin-based regimens are safe in CKD (including people on dialysis):[5]
• atorvastatin 20 mg
• rosuvastatin 10 mg
• simvastatin 20 mg combined with ezetimibe 10 mg</td></tr>
</table>

Generic Name *Trade name* Dosage form and strengths	**Recommended Renoprotective Dose**[A]	**Approx. Cost per year**[D]	**PharmaCare Coverage**[C]	**Adverse Effects**[B]	**Drug Interactions**[B] **and Therapeutic Considerations for Chronic Kidney Disease**
Glucagon-like Peptide-1 Receptor Agonists (GLP-1 RA) Examples listed below. See BCGuidelines: Diabetes Care Antihyperglycemic drug table for more info.					
Semaglutide *Ozempic* Multi-dose pre-filled pens delivering doses of 0.25, 0.5, 1,2mg	**Initial:** 0.25 mg subcut once weekly x 4 weeks, then increase to 0.5 mg subcut weekly. May increase to 1 mg subcut weekly after additional 4 weeks **Usual:** 0.5 – 1 mg subcut once weekly **Max:** 1 mg subcut once weekly (CKD trial dose)[8]	$3,065	Limited coverage for treatment of type 2 diabetes	Common: • Nausea • Vomiting • Diarrhea • Constipation • Gastroparesis Uncommon: • Pancreatitis • Bowel obstruction • Cholelithiasis	Prioritize using agents that do not require dose adjustment in CKD and have documented cardiovascular, kidney, and mortality benefits:[5] • semaglutide (injectable)* • liraglutide • dulaglutide *Note: Only semaglutide has been studied in a CKD population.[5,8] Benefits of other GLP-1 RA's are extrapolated from trials in people with type 2 diabetes where kidney function was generally preserved.[5]
Dulaglutide *Trulicity* Single-use pre-filled pen delivering doses of 0.7, 1.5 mg	**Initial:** 0.75 mg subcut weekly; may increase to 1.5 mg subcut weekly after 1 week Titrate every 4 weeks **Max:** 4.5 mg subcut weekly	$9,830	Non-benefit		
Liraglutide *Victoza* Multi-use pre-filled pen delivering doses of 0.6, 1.2, 1.8 mg	**Initial:** 0.6 mg subcut daily for 1 week; then increase to 1.2 mg subcut daily **Max:** 1.8 mg subcut daily	$4,090	Non-benefit		
Dihydropyridine Calcium Channel Blockers (e.g., amlodipine) See BCGuidelines: Hypertension Antihypertensive drug table for more info.					
Diuretics See BCGuidelines: Hypertension Antihypertensive drug table and BCGuidelines: Heart Failure drug table for more info.					
In randomized controlled trials of primary hypertension examining the effect of antihypertensive drugs on cardiovascular outcomes that included participants with CKD, cardiovascular benefits have been most consistent with thiazide-like diuretics (e.g., hydrochlorothiazide, chlorthalidone, etc.).[9] There are no data on clinical outcomes with loop diuretics (e.g., furosemide) in the treatment of high blood pressure with or without CKD.[9]				• Chlorthalidone, indapamide, and metolazone appear to remain effective for blood pressure lowering and/or diuresis at GFRs < 30 ml/min per $1.73m^2$.[9] • Loop diuretics are effective at lower GFRs (i.e., < 30 ml/min per 1.73 m^2).[9] • Withhold during times of prolonged fasting, critical illness, acute management of adverse effects, or 48-72 hours prior to surgery; **must have a plan to restart**.[5]	

Generic Name *Trade name* Dosage form and strengths	Recommended Renoprotective Dose[A]	Approx. Cost per year[D]	PharmaCare Coverage[C]	Adverse Effects[B]	Drug Interactions[B] and Therapeutic Considerations for Chronic Kidney Disease
Nonsteroidal Mineralocorticoid Receptor Antagonists (ns-MRA)					
Finerenone *Kerendia* Tabs: 10, 20 mg	10 mg PO once daily if eGFR 25 - 59 ml/min/1.73m^2 20 mg PO once daily if eGFR ≥ 60 ml/min/1.73m^2	$1,285	Limited coverage as an adjunct to standard care therapy in adults with both CKD and type 2 diabetes	Common: • Hyperkalemia (hospitalization for serious hyperkalemia rare)[5] Less common: • Hypotension • Hyponatremia • Anemia	• Contraindications: - Concomitant strong CYP3A1 inhibitors (e.g., itraconazole, ketoconazole, ritonavir, clarithromycin, etc.)[10] - Addison's disease[10] • Initiation not recommended in patients with eGFR < 25 mL/min/1.73m^2 as clinical experience is limited.[5] • Mitigate risk of hyperkalemia by selecting people with consistently normal serum potassium concentration and monitor serum potassium regularly after initiation.[5] • Monitor potassium 1 month after initiation then every 4 months.[5] • Hold if potassium > 5.5 mmol/l; consider changes to diet or medications to mitigate hyperkalemia; consider reinitiation if / when potassium ≤ 5.0 mmol/l.[5]
Steroidal Mineralocorticoid Receptor Antagonists (MRA) Example listed below. See BCGuidelines: Heart Failure drug table for more info.					
Spironolactone *Aldactone, G* Tabs: 25, 100 mg	**Initial:** 12.5 mg PO once daily **Target:** 25 – 50 mg daily	$20	Regular benefit	• Hyperkalemia • Dehydration • Nausea • Gynecomastia (usually reversible upon discontinuation)	• Avoid in people with high risk of hyperkalemia (e.g., hypoaldosteronism or type 4 renal tubular acidosis).[9] • Concomitant use with ACE-I or ARB increases risk of hyperkalemia and decline in kidney function, especially among people with eGFR < 45 ml/min per 1.73m^2.[9]

Generic Name *Trade name* Dosage form and strengths	**Recommended Renoprotective Dose**[A]	**Approx. Cost per year**[D]	**PharmaCare Coverage**[C]	**Adverse Effects**[B]	**Drug Interactions**[B] **and Therapeutic Considerations for Chronic Kidney Disease**
Treatments for Complications Attributable to Chronic Kidney Disease					
Metabolic Acidosis					
Sodium bicarbonate *G* Tabs: 325 mg (3.8 mmol bicarbonate), 500 mg (5.8 mmol bicarbonate), 1000 mg Liquid: 7.5% Injectable: 4.2%, 8.4%	**Initial:** 325–500 mg PO BID or TID. Titrate to achieve bicarbonate level > 18 mmol/l and within the expected range. **Max:** 5850 mg/day	$83 (500 mg TID)	Non-benefit	• Bloating, flatulence, increased sodium absorption • Reduced absorption of medications requiring an acidic gastric pH, e.g., atazanavir, calcium carbonate, iron tablets, itraconazole, ketoconazole	• Consider use of pharmacological treatment with or without dietary intervention to prevent development of acidosis with potential clinical implications (e.g., serum bicarbonate < 18 mmol/l in adults).[5] • Monitor for fluid retention and heart failure.[12] • Monitor for metabolic acidosis to ensure it does not result in serum bicarbonate concentrations exceeding the upper limit of normal and does not adversely affect BP control, serum potassium, or fluid status.[5] • Avoid serum bicarbonate > 32 mmol/L as it is associated with increased mortality in patients with CKD.[12] • Baking soda dissolved in water may be used as an alternative to sodium bicarbonate in patients who cannot take tablets (⅛ teaspoon = 7.1 mmol bicarbonate).[12]
Citric acid/ sodium citrate *(Shohl's solution)* *Dicitrate, G* Liquid: 1 mmol bicarbonate/ml	**Initial:** 0.5 mmol/kg/ day in 2–3 divided doses **Target:** Titrate to achieve bicarbonate level > 18 mmol/l and within the expected range.	$869 (0.5 mmol/ 70 kg/ day)	Non-benefit		
Iron Deficiency Anemia See BCGuidelines: Iron Deficiency oral iron formulations and dosages table for more info.					
Oral iron products • Ferrous sulfate • Ferrous gluconate • Ferrous fumarate • Polysaccharide iron • Heme iron polypeptide	UpToDate recommends starting with ferrous sulfate 325mg (65 mg elemental iron) to reach either of the following dosing regimens and goals: [11] • Daily dosing goals: elemental iron approx. 200 mg per day in up to three divided doses • Alternate-day dosing goal: elemental iron approx. 65 mg per day in a single dose		Regular benefit: ferrous sulfate, gluconate, fumarate Non-benefit: polysaccharide iron, heme iron polypeptide	• Although ferrous sulfate is commonly available and inexpensive, other oral iron preparations may also be used; there is not significant evidence to suggest that other oral iron formulations are more effective or associated with fewer adverse side effects than ferrous sulfate.[13] • Alternate-day dosing has not been tested in people with CKD. Despite lack of data, it is reasonable to choose this regimen due to potential benefits with regards to iron absorption and side effects.[11]	
Potassium Abnormalities					

Consider potassium exchange agents as a part of second-line therapy to manage serum potassium > 5.5 mmol/l in CKD.[5]

1st line: **Address correctable factors**	• Review non-RASi medications (e.g. NSAIDs, trimethoprim) • Assess dietary potassium intake (dietary referral) and consider appropriate moderation of dietary potassium intake
2nd line: **Medications**	Consider: • Appropriate use of diuretics • Optimize serum bicarbonate levels • Licensed potassium exchange agents
3rd line: **Last resort**	• Reduce dose or discontinue RASi/MRA (Discontinuation is associated with increased cardiovascular events. Review and restart RASi or MRA at a later date if patient condition allows.)

Figure 32 | Actions to manage hyperkalemia (potassium > 5.5 mmol/l) in chronic kidney disease. MRA, mineralocorticoid receptor antagonists; NSAID, nonsteroidal anti-inflammatory drug; RASi, renin-angiotensin system inhibitors.

Abbreviations: **ACE-I** ACE inhibitor; **ARB** angiotensin receptor blockers; **BID** twice a day; **Cap** capsules; **CKD** chronic kidney disease; **CR** controlled release; **G** generics; **IR** immediate release; **IV** intravenous; **ODT** oral dissolving tablet; **LA** long acting; **SR** sustained release; **Tab** tablets; **XR** extended release

[A] Dosages used in CKD trials, if available. Consult product monograph for detailed dosing instructions and dose adjustments for unique patient populations.

[B] Not an exhaustive list. Check the product monograph (https://health-products.canada.ca/dpd-bdpp/index-eng.jsp) or an interaction checker (e.g., Lexicomp(c)) before prescribing

[C] PharmaCare coverage as of October 2024 (subject to revision). Regular Benefit: Eligible for full reimbursement.* Limited Coverage: Requires Special Authority to be eligible for reimbursement.* Reference Drug Program (RDP): may be regular or partial benefit.* Non-benefit: Not eligible for reimbursement. *Reimbursement is subject to the rules of a patient's PharmaCare plan, including any deductibles. In all cases above, coverage is subject to drug price limits set by PharmaCare. See: https://www.gov.bc.ca/pharmacareplans and https://www.gov.bc.ca/pharmacarepolicy for further information.

[D] Drugs costs are average retail cost of the generic, when available. Costs are for maximum dosages where dosage ranges are provided. Current as of October 2024 and does not include retail markups or pharmacy fees.

[E] BC Renal: provides coverage for registered patients who meet criteria. See http://www.bcrenal.ca/health-professionals/clinical-resources/pharmacy-formulary for further information.

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